Einstein–Maxwell Waves inside Weyl–Maxwell Gravity on a Compact Product: Exact Linear Phase-Space Inclusion and the Extra Axial Branch

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Abstract

We locate ordinary Einstein–Maxwell gravitational waves inside pure Weyl–Maxwell gravity on the compact product $\mathbb{R}_t \times S^1_L \times S^2$. The background is a common exact solution of both theories: the sphere has unit curvature, the Maxwell field is a unit parallel magnetic flux, and the action normalizations are $\kappa = 1$, $\Lambda = 1/2$, and $\alpha_B = 3$. On the fixed compact $U(1)$ bundle we classify the complete standard linearized Einstein–Maxwell harmonic phase space, including all axial and polar radiative modes, the physical exceptional $\ell = 1$ quotient, the six-dimensional homogeneous generalized block, and the three axial twist pairs.

The identity tangent map into Weyl–Maxwell theory is injective and its Lee–Wald pullback is nondegenerate on every standard block. It is not a symplectic inclusion. For $\ell \geq 2$ the target form is the Einstein–Maxwell form composed with the parity-independent spectral polynomial

$$p_\lambda(M) = 1 + \frac{3}{2}(M - \lambda),$$

whose two normal-mode weights are $1 \pm \frac{3}{2}\sqrt{2\lambda}$. Thus each real spatial harmonic has two positive and two negative relative branch coefficients. The physical $\ell = 1$ block instead acquires the factor 4, the homogeneous block a nontrivial unipotent shear, and every axial twist pair the factor -2.

We then solve the complete generic axial Weyl–Maxwell target. Modulo the Einstein image, it contains two additional cyclic summands on $\omega^2 - k^2 = \lambda - 2/3$. Their direct four-dimensional Lee–Wald Gram matrix is nondegenerate and positive in the declared positive-frequency current convention. The Einstein and extra modules are mutually orthogonal, and the normalized positive-frequency current has axial inertia $(3, 1)$. Inverting the extra Gram matrix gives two conserved spectral coefficient extractors which annihilate the Einstein image.

These are classical LOCAL-ALGEBRAIC/REDUCED-MODE statements before the final residual quotient. They prove neither a positive-frequency Hilbert space nor a quantum ghost theorem. They also do not construct asymptotically flat scattering. The ordinary waves are present and nonnull before the global residual reduction; a vanishing one-particle residual cohomology on a closed cylinder therefore answers a different question from the existence of local gravitational radiation.

AI-use disclosure and computational accountability

OpenAI GPT-5 was used substantively for mathematical synthesis, symbolic-code generation and debugging, literature organization, claim auditing, and manuscript drafting. The human author

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posed and refined the scientific questions, directed the repository workflow, and selected and reviewed the claims and derivations in this draft. The human author retains responsibility for the final text, and submission remains conditional on a documented final human review. Every algebraic result used below is reproduced by deterministic exact scripts and checked by a separately implemented deterministic verifier that does not import the generator code; Section 9 gives content hashes and commands. No floating-point calculation is used in a rank, inertia, quotient, or on-shell identity. This disclosure follows the current APS requirement that substantive AI use be disclosed while the AI program is not listed as an author [17].

Contents

1 The question and the answer

An Einstein–Maxwell solution need not solve Weyl–Maxwell theory, and even when it does, its tangent solutions need not inherit the Einstein–Maxwell phase-space form. We study a magnetically supported compactification of the Plebański–Hacyan direct-product electrovacuum [2, 8, 9]. It lies on the intersection of the two theories, and we determine both its induced Einstein image and the additional axial target. The relevant question is not whether one can fix Weyl gauge to obtain Einstein gravity—one cannot do so in general [1, 14, 15]—but instead:

On a common Einstein–Maxwell/Weyl–Maxwell background, which ordinary wave solutions embed, what phase-space form do they inherit, and which additional Weyl–Maxwell solutions remain?

To our knowledge, this paper answers that question for the first time as one combined phase-space/current/module calculation: completely for the standard harmonic Einstein–Maxwell tangent and for the generic axial part of the additional target on a fixed compact magnetic bundle. The answer has three logically separate layers:

1. every standard Einstein–Maxwell tangent class injects as a Weyl–Maxwell solution class;
2. the target Lee–Wald form restricts nondegenerately to that image, but does not equal the source form under the identity map; and
3. the generic axial Weyl–Maxwell target contains two further nonradical solution directions which are not Einstein representatives or local gauge.

Let $\mathcal{T}^{\rm std}_{\rm EM}$ denote the complete fixed-bundle standard Einstein–Maxwell harmonic tangent after local Diff $\times U(1)$ reduction, and let $\mathcal{T}^{\rm ax}_{\rm WM}$ denote the generic axial Weyl–Maxwell tangent after local Diff $\times$Weyl $\times\, U(1)$ reduction. The central result can be stated schematically as

$$\boxed{\begin{gathered}\iota : \mathcal{T}^{\rm std}_{\rm EM} \hookrightarrow \mathcal{T}_{\rm WM}, \qquad \ker(\iota^*\Omega_{\rm WM}) = 0,\\ \iota^*\Omega_{\rm WM} \neq \Omega_{\rm EM},\\ (\mathcal{T}^{\rm ax}_{\rm WM}/\iota(\mathcal{T}^{\rm ax}_{\rm EM}))_{\ell,n} \cong \big(K_{\ell,n}[\omega]/(p)\big)^2, \qquad p = \omega^2 - k^2 - \lambda + \frac{2}{3}.\end{gathered}} \tag{1}$$

Here $K_{\ell,n}$ is the characteristic-zero coefficient field after the physical specialization $\lambda = \ell(\ell+1)$ and $k = 2\pi n/L$. The last line holds for every $\ell \geq 2$ and every $n \in \mathbb{Z}$, including $n = 0$; it is a canonical fibrewise module quotient, not a choice of complement. Section 5 proves this without inverting k, ω, or either shell polynomial.

1.1 What is and is not frozen

The theorem level of this manuscript is deliberately asymmetric in parity. The standard Einstein–Maxwell image is complete in both axial and polar parities and includes all exceptional/global blocks. The additional Weyl–Maxwell branch is complete only in the generic axial sector. The polar extra branch and the final residual descent remain open. Accordingly the paper makes no parity-complete claim about the full Weyl–Maxwell target.

The dependency labels are

LOCAL-ALGEBRAIC and REDUCED-MODE.

No result below carries a LORENTZIAN-CAUSAL dependency label. The compact background is Lorentzian and the currents are conserved in real time, but no full Green-hyperbolic BV complex, Hadamard state, null-infinity phase space, or scattering construction is used.

2 The two theories and their common compact background

We use signature $(-,+,+,+)$ and the actions

$$S_{\mathrm{EM}}[g,A]=\int_M\sqrt{-g}\left[\frac{R-2\Lambda}{2\kappa}-\frac{1}{4}F_{ab}F^{ab}\right]d^4x, \tag{2}$$

$$S_{\mathrm{WM}}[g,A]=\int_M\sqrt{-g}\left[\frac{\alpha_B}{8}C_{abcd}C^{abcd}-\frac{1}{4}F_{ab}F^{ab}\right]d^4x. \tag{3}$$

Our Bach convention is fixed by the metric variation

$$\delta\int_M\sqrt{-g}\,C_{abcd}C^{abcd}\,d^4x=4\int_M\sqrt{-g}\,B_{ab}\,\delta g^{ab}\,d^4x+\int_{\partial M}\theta_C(g;\delta g), \tag{4}$$

whereas $\delta S_{\mathrm{Maxwell}}=-\frac{1}{2}\int\sqrt{-g}\,T_{ab}\delta g^{ab}$ up to its boundary potential. Thus the bulk variation of (3) is $\frac{1}{2}\int\sqrt{-g}(\alpha_B B_{ab}-T_{ab})\delta g^{ab}$, which fixes both the factor and sign in the field equation below. Thus

$$G_{ab}+\Lambda g_{ab}=\kappa T_{ab},\qquad \nabla_a F^{ab}=0, \tag{5}$$

$$\alpha_B B_{ab}=T_{ab},\qquad \nabla_a F^{ab}=0, \tag{6}$$

with $dF=0$ and

$$T_{ab}=F_{ac}F_b{}^c-\frac{1}{4}g_{ab}F_{cd}F^{cd}. \tag{7}$$

Consider a product $M_2(k_1)\times\Sigma_2(k_2)$ of oriented constant curvature surfaces and an aligned parallel field

$$F=E\ \mathrm{vol}(M_2)+P\ \mathrm{vol}(\Sigma_2). \tag{8}$$

The same metric and field solve both theories on the incidence branch

$$\Lambda=\frac{k_1+k_2}{2},\qquad E^2+P^2=\frac{k_2-k_1}{\kappa},\qquad \alpha_B\kappa(k_1+k_2)=3. \tag{9}$$

This is an intersection of solution loci, not a gauge equivalence.

We specialize to the positive flat branch, locally the magnetically supported Plebański–Hacyan universe and globally its spatial circle compactification,

$$M=\mathbb{R}_t\times S^1_L\times S^2,\qquad (k_1,k_2)=(0,1),\qquad (\kappa,\Lambda,\alpha_B,E,P)=(1,\tfrac{1}{2},3,0,1). \tag{10}$$

The spatial Cauchy surface is the closed manifold $S^1\times S^2$. With minimal charge $q_{\mathrm{min}}=1$, flux quantization gives Chern number $N=2$. Throughout the paper the bundle P_N is fixed. Uniform magnetic variation would change the topological sector and is not an allowed tangent. The local direct-product background belongs to the family studied in the Einstein–Maxwell literature [2, 8, 9] and in earlier and recent Bach/Weyl–Maxwell product analyses [3, 10]. The novelty claimed here is not the existence of the product or of selected exact waves. It is the complete fixed-bundle standard harmonic phase space, its exact relative Lee–Wald endomorphism, and the nonradical generic axial quotient.

2.1 Connections, bundle automorphisms, and global zero modes

Because $c_1(P_N)\neq 0$, the background potential $\bar{A}$ is a connection on P_N, not a global one-form. For a second connection A on the same bundle, however,

$$a=A-\bar{A}\in\Omega^1(M) \tag{11}$$

is a global adjoint-valued one-form. A bundle-covariant infinitesimal lift of a diffeomorphism and a vertical gauge transformation acts by

$$\delta_{\xi,\chi}h = \mathcal{L}_\xi \bar{g}, \qquad \delta_{\xi,\chi}a = \iota_\xi \bar{F} + d\chi. \tag{12}$$

Locally this is the familiar $\mathcal{L}_\xi \bar{A} + d\chi_{\text{local}}$ after absorbing $\iota_\xi \bar{A}$ into the vertical parameter. Equation (12), rather than a globally defined $\bar{A}$, is used when subtracting the common Diff $\ltimes \mathcal{G}_{U(1)}$ action in the injectivity argument below.

At the infinitesimal identity-component level the constant circle component W_x is a real tangent coordinate. Globally, if x has period L, a large $U(1)$ transformation shifts

$$W_x \sim W_x + \frac{2\pi m}{q_{\min} L}, \qquad m \in \mathbb{Z}. \tag{13}$$

Thus the finite holonomy is a circle coordinate, while Q_e is its local cotangent partner. Similarly, the three constant axial twists describe the tangent to an $SO(3)$ isometry holonomy around S^1. Large diffeomorphisms and conjugation can identify finite points; our theorem concerns the unreduced local tangent before that global moduli quotient.

2.2 Why the complete linear inclusion holds

To remove a convention ambiguity, let

$$\begin{aligned} \mathsf{S}_{abcd} = & -\nabla_a F_{ce} \nabla_b F_d{}^e - \nabla_a F_{de} \nabla_b F_c{}^e + g_{ab} \nabla_f F_{ce} \nabla^f F_d{}^e \\ & + \frac{1}{2} g_{cd} \nabla_a F_{ef} \nabla_b F^{ef} - \frac{1}{4} g_{ab} g_{cd} \nabla_e F_{fh} \nabla^e F^{fh}, \\ H_{abcd} := & \frac{1}{2}(\mathsf{S}_{abcd} + \mathsf{S}_{cdab}), qquad H_{ab} := H_{abc}{}^c, qquad C^{\text{Ch}}_{ab} := 2\kappa H_{ab}. \end{aligned} \tag{14}$$

Thus C^{Ch} is explicitly homogeneous quadratic in ∇F; the factor 2κ is part of our convention rather than an unstated change of the Chevreton tensor. For a source-free four-dimensional Einstein–Maxwell solution with cosmological constant, the convention-adjusted trace of the Chevreton tensor obeys the exact on-shell identity

$$B_{ab} - \frac{2\kappa\Lambda}{3} T_{ab} = C^{\text{Ch}}_{ab}. \tag{15}$$

This is Eq. (66) of Bergqvist and Eriksson translated to (4), with the nonzero cosmological term retained [5, 6].

An on-shell identity cannot be differentiated only along actual nonlinear solution families if one wishes to cover arbitrary Jacobi fields. The following formal bridge supplies the missing step.

Proposition 2.1 (Dual-number formal linearization)**.** *Let (h, f) be any smooth fixed-bundle solution of the complete linearized Einstein, Maxwell, and Bianchi equations at $(\bar{g}, \bar{F})$. Then*

$$D\left(B_{ab} - \frac{2\kappa\Lambda}{3} T_{ab} - C^{\text{Ch}}_{ab}\right)_{(\bar{g},\bar{F})} [h, f] = 0. \tag{16}$$

No assumption that (h, f) integrates to a real one-parameter family is required.

Proof. Work in the dual-number algebra $\mathbb{D} = \mathbb{R}[\varepsilon]/(\varepsilon^2)$ and set $(g_\varepsilon, F_\varepsilon) = (\bar{g} + \varepsilon h, \bar{F} + \varepsilon f)$. The metric is invertible over $\mathbb{D}$, with

$$g_\varepsilon^{-1} = \bar{g}^{-1} - \varepsilon \bar{g}^{-1} h \bar{g}^{-1}. \tag{17}$$

Because the zeroth-order fields solve the nonlinear equations and (h, f) solves their linearizations, the full Einstein, Maxwell, and Bianchi residuals of $(g_\varepsilon, F_\varepsilon)$ vanish in $\mathbb{D}$.

Now repeat the derivation of (15) over $\mathbb{D}$. Its first step is the Leibniz expansion of H_{ab}; the Maxwell wave identity then follows by differentiating $dF = 0$ and $\nabla^a F_{ab} = 0$ and commuting covariant derivatives. The Ricci terms are replaced using the Einstein equation, and the remaining curvature derivatives are converted by the differential Bianchi identity and the geometric definition of B_{ab}. These operations are natural tensor algebra, covariant differentiation, and curvature commutation, so they remain valid over $\mathbb{D}$. Hence (15) holds modulo ε^2 for $(g_\varepsilon, F_\varepsilon)$; its ε coefficient is (16). This repeats the on-shell derivation at first formal order rather than assuming nonlinear integrability. □

Consequently, on the formal Einstein–Maxwell shell,

$$\alpha_B B_{ab} - T_{ab} = \alpha_B C^{\mathrm{Ch}}_{ab} + \left(\frac{2\alpha_B \kappa \Lambda}{3} - 1\right) T_{ab}. \tag{18}$$

The incidence relation gives $2\alpha_B\kappa\Lambda/3 = 1$. On the product fixture $\bar{\nabla}\bar{F} = 0$, while C^{Ch} is homogeneous quadratic in ∇F. Therefore

$$C^{\mathrm{Ch}}(\bar{g}, \bar{F}) = 0, \qquad \delta C^{\mathrm{Ch}}_{(\bar{g},\bar{F})}[h, a] = 0, \tag{19}$$

and linearizing (18) along any solution of the complete linearized Einstein–Maxwell equations gives $\alpha_B \delta B_{ab} - \delta T_{ab} = 0$. The Maxwell equation is identical in the two theories, so the same pair (h, a) solves the complete linearized Weyl–Maxwell equations.

Proposition 2.1 covers all formal linearized solutions, including non-integrable ones. It is nevertheless weaker than an explicit off-shell factorization by global Euler–Lagrange row operators or an off-shell BV chain map. The Chevreton structure also predicts its own limitation: the first possible Chevreton failure is quadratic, so no nonlinear closure follows from the linear theorem.

Let

$$\iota[(h, a)]_{\mathrm{Diff}\,\times U(1)} = [(h, a)]_{\mathrm{Diff}\,\times \mathrm{Weyl} \times U(1)}. \tag{20}$$

The map is well defined on the declared linear solution quotients. It is also injective. Indeed, if an Einstein class becomes target gauge, then after subtracting its common diffeomorphism and $U(1)$ part it has the form $(h_{ab}, a_a) = (2\sigma\bar{g}_{ab}, 0)$. The linearized Einstein–Maxwell rows imply

$$-3\Delta_{S^2}\sigma + 2\sigma = 0. \tag{21}$$

On $Y_{\ell m}$ this is multiplication by $3\ell(\ell + 1) + 2$, hence smoothness forces $\sigma = 0$.

Theorem 2.2 (Fixed-bundle tangent inclusion)**.** *On the fixed background* (10)*, the identity tangent map sends every complete linearized Einstein–Maxwell solution to a linearized Weyl–Maxwell solution and descends injectively through the respective local gauge quotients.*

Remark 2.3. Theorem 2.2 does not say that the two theories are gauge equivalent. In particular, a generic Bach-flat solution need not be conformally Einstein, and Section 5 constructs additional target directions explicitly.

3 The complete standard Einstein–Maxwell harmonic phase space

Write

$$\lambda = \ell(\ell + 1), \qquad k = k_n = \frac{2\pi n}{L}, \qquad -\Delta_{S^2} Y_{\ell m} = \lambda Y_{\ell m}, \tag{22}$$

and $N_{\ell m} = \int_{S^2} |Y_{\ell m}|^2 d\Omega > 0$. A complex coefficient at (n, ℓ, m) is paired by reality with the coefficient at $(-n, \ell, -m)$. Equivalently one may use real spherical harmonics and cosine/sine circle modes. We use the latter convention for real mode counts. The construction is complementary to the standard gauge-invariant Einstein–Maxwell master-equation framework [7]: the compact product, fixed magnetic bundle, exceptional harmonics, and global Jordan blocks require the separate reduction given here.

3.1 Regular radiative blocks

For $\ell \geq 2$, the axial and polar reductions each have two masters. In the orders (H, Q) and (K, U) their normal-mode matrices are

$$M_A = \begin{pmatrix} \lambda & 2 \\ \lambda & \lambda \end{pmatrix}, \qquad M_P = \begin{pmatrix} \lambda & -2\lambda \\ -1 & \lambda \end{pmatrix}. \tag{23}$$

Both parities have the exact dispersions

$$\omega_\pm^2 = k^2 + \lambda \pm \sqrt{2\lambda}. \tag{24}$$

The source action-normalized Wronskians are controlled, up to the common positive factor $N_{\ell m}/2$, by

$$G_A = \begin{pmatrix} \lambda^2 & 2\lambda \\ 2\lambda & 2\lambda \end{pmatrix}, \qquad G_P = \begin{pmatrix} 1 & -2 \\ -2 & 2\lambda \end{pmatrix}. \tag{25}$$

They obey $G_A M_A = M_A^T G_A$ and $G_P M_P = M_P^T G_P$. Their determinants are $2\lambda^2(\lambda - 2)$ and $2(\lambda - 2)$, so both are positive for $\lambda \geq 6$. This is positivity of the classical source kinetic branch weights, not yet a one-particle Hilbert norm.

The branch eigenvectors may be taken as

$$v_{A,\pm} = (1, \ \pm\sqrt{\lambda/2}), \qquad v_{P,\pm} = (\mp\sqrt{2\lambda}, \ 1). \tag{26}$$

These compact parity masters are the exact representation of the ordinary coupled photon/graviton-like waves. On the magnetically supported product the metric and Maxwell coefficients mix; helicity ± 2 is therefore a local/high-frequency interpretation, whereas (n, ℓ, m), parity, and branch are the exact global labels.

3.2 Exceptional and global blocks

At $\ell = 1$ one branch becomes gauge in each parity. The axial source matrix has kernel $(H, Q) = (1, -1)$ and physical direction $(1, 1)$. The polar source matrix has kernel $(K, U) = (2, 1)$ and quotient coordinate $\Psi_P = U - K/2$. The surviving axial and polar modes obey $\omega^2 = k^2 + 4$. They are radiative triplets, not the zero-frequency twist.

The complete homogeneous generalized block is six-dimensional. A convenient representative is

$$K = a + bt, \qquad C = at^2 + \frac{b}{3}t^3 + c + dt, \qquad A_x = W_x + Q_e t. \tag{27}$$

Polynomial Jordan partners are retained; imposing boundedness in time would define a different phase space. After the common factor $2\pi L$ is removed, the source form in the order (a, b, c, d, Q_e, W_x)

is

$$\Omega^{(0)}_{\mathrm{EM}} = \begin{pmatrix} 0 & -1 & 0 & -1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 & 2 & 0 \end{pmatrix}. \tag{28}$$

It has rank six. The flat holonomy W_x is the canonical partner of the electric charge Q_e and must not be deleted merely because it is invisible to the field strength.

For every one of the three real axial $\ell = 1$ harmonics, the generalized twist

$$h_{xa} = (A_m + B_m t)X_a, \qquad a_x = -(A_m + B_m t)Y_{1m} \tag{29}$$

forms a Darboux pair. The would-be gauge generator is proportional to $x(A_m + B_m t)$ and is not periodic on S^1.

For completeness, Table 1 records every exceptional intersection. "Equation reduction" counts the surviving coefficient functions after gauge fixing and then the independent second-order masters; it is more informative here than the rank of one Fourier matrix, whose rank jumps at the generalized zero-frequency block. Phase dimensions and current ranks are real and are stated per real spatial harmonic.

Stratum	raw coefficients	gauge rank	equation reduction	$\dim \mathcal{T}$	$\operatorname{rank} \Omega_{\mathrm{EM}}$
$\ell = 0$, $k \neq 0$	5	2	$3 \to 0$	0	0
$\ell = 0$, $k = 0$ generalized	$5 + W_x$	2	(a, b, c, d, Q_e, W_x)	6	6
$\ell = 1$, $k \neq 0$	$5_{\mathrm{ax}} + 7_{\mathrm{pol}}$	$2 + 3$	$3 + 4 \to 1 + 1$	4	4
$\ell = 1$, $k = 0$	$5_{\mathrm{ax}} + 7_{\mathrm{pol}}$	$2 + 3$	$1 + 1$ masters + twist	6	6
$\ell \geq 2$, $k = 0$	$6_{\mathrm{ax}} + 8_{\mathrm{pol}}$	$2 + 3$	$4 + 5 \to 2 + 2$	8	8
$\ell \geq 2$, $k \neq 0$	$6_{\mathrm{ax}} + 8_{\mathrm{pol}}$	$2 + 3$	$4 + 5 \to 2 + 2$	8	8

Table 1: Complete standard source strata before the final residual quotient. At $\ell = 0$, every nonzero Fourier block is pure gauge and the electric coefficient vanishes; exact rank witnesses are $k^4/2$ for $k \neq 0$ and $\omega^4/2$ for $k = 0$, $\omega \neq 0$. The displayed homogeneous row keeps the polynomial Jordan solutions rather than imposing temporal boundedness.

Theorem 3.1 (Complete source phase space)**.** *Before the final residual quotient, the fixed-bundle standard harmonic Einstein–Maxwell tangent is the symplectic direct sum*

$$\mathcal{T}^{\mathrm{std}}_{\mathrm{EM}} = \mathcal{T}^{\ell \geq 2}_{\mathrm{rad}} \oplus \mathcal{T}^{\ell=1}_{\mathrm{phys}} \oplus \mathcal{T}^{\ell=0}_{\mathrm{hom}} \oplus \mathcal{T}^{\ell=1,\omega=0}_{\mathrm{twist}}. \tag{30}$$

The regular radiative forms, exceptional physical quotient, homogeneous block, and three twist pairs are all nondegenerate.

4 The relative phase-space endomorphism on the Einstein image

A solution inclusion need not preserve a symplectic structure. We use the covariant phase-space current in the Lee–Wald convention [11, 12] and therefore keep three objects separate:

$$\Omega_{\mathrm{EM}}, \qquad \iota^*\Omega_{\mathrm{WM}}, \qquad \Omega_{\mathrm{WM}} \text{ on the full target.} \tag{31}$$

The first two live on the same vector space but arise from different actions. Because Ω_{EM} is nondegenerate on the source phase space, define the relative endomorphism R invariantly by

$$\iota^*\Omega_{\mathrm{WM}}(u,v) = \Omega_{\mathrm{EM}}(u,Rv). \tag{32}$$

The conjugacy class of R, with the identity field map and source normal-mode identification fixed, is the comparison datum. Abstractly, any two finite-dimensional nondegenerate real symplectic spaces of the same dimension are symplectomorphic; our claim is therefore not an absolute obstruction to some unrelated change of phase-space coordinates.

4.1 Regular radiation: one spectral polynomial

On both parities and for every $\ell \geq 2$,

$$\iota^*\Omega_{\mathrm{WM}}(u,v) = \Omega_{\mathrm{EM}}\big(u, p_\lambda(M)v\big), \qquad p_\lambda(x) = 1 + \frac{3}{2}(x-\lambda). \tag{33}$$

In the axial and polar master coordinates,

$$p_\lambda(M_A) = \begin{pmatrix} 1 & 3 \\ 3\lambda/2 & 1 \end{pmatrix}, \qquad p_\lambda(M_P) = \begin{pmatrix} 1 & -3\lambda \\ -3/2 & 1 \end{pmatrix}. \tag{34}$$

On the normal branches its eigenvalues are

$$r_+ = 1 + \frac{3}{2}\sqrt{2\lambda} > 0, \qquad r_- = 1 - \frac{3}{2}\sqrt{2\lambda} < 0. \tag{35}$$

Neither vanishes for $\lambda \geq 6$. The axial and polar branch weights are identical even though their off-shell coefficient matrices differ.

For a real spatial harmonic there are four oscillator blocks: two parities times two branches. Hence the inertia of the symmetric branch-coefficient form associated with R is

$$\mathrm{inertia}_{\mathrm{rel}}(R) = (2,2). \tag{36}$$

This counts the signs of the symmetric coefficients multiplying the standard real Darboux blocks. It is not a Sylvester signature of a real symplectic form, which has no such invariant.

4.2 Exceptional and global restrictions

The exceptional $\ell = 1$ calculation cannot be obtained by blindly setting $\lambda = 2$ in the generic polar matrix: that continuation fails target gauge descent. Direct four-dimensional currents instead give, in source-normalized axial and polar quotient coordinates,

$$\iota^*\Omega_{\mathrm{WM}}|_{\ell=1,\mathrm{phys}} = 4\;\Omega_{\mathrm{EM}}|_{\ell=1,\mathrm{phys}}\,. \tag{37}$$

The literal target current has zero gauge rows and columns, so this is a quotient identity rather than only a norm comparison.

On the homogeneous block, after the same common factor as in (28),

$$\Omega^{(0)}_{\mathrm{WM}} = \begin{pmatrix} 0 & 2 & 0 & -1 & 0 & 0 \\ -2 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 & 2 & 0 \end{pmatrix}. \tag{38}$$

In the matrix convention of (32), one obtains

$$R = I + N, \qquad N^2 = 0, \qquad \operatorname{rank} N = 2. \tag{39}$$

Thus the identity is not symplectic, although $S = I + N/2$ has determinant one and satisfies $S^T \Omega_{\mathrm{EM}}^{(0)} S = \Omega_{\mathrm{WM}}^{(0)}$.

Finally, each generalized axial twist pair obeys the direct identity

$$\iota^* \Omega_{\mathrm{WM}}|_{\mathrm{twist}} = -2\, \Omega_{\mathrm{EM}}|_{\mathrm{twist}}\,. \tag{40}$$

It comes from the full $A + Bt$ current and is not a zero-frequency limit of an oscillatory formula.

4.3 Orthogonality and the assembled theorem

Different circle momenta, spherical irreducibles, and parity sectors are orthogonal by S^1 Fourier invariance, $SO(3)$ invariance, and spatial parity. Distinct radiative branches are orthogonal because M is Ω_{EM}-self-adjoint and has distinct eigenvalues. The only shared label not settled by these arguments is the physical axial $\ell = 1, n = 0$ oscillator against the twist. Writing p_{osc} for the arbitrary physical-oscillator amplitude, its literal mixed target current is proportional to

$$-2i\pi\omega p_{\mathrm{osc}}(\omega^2 - 4)[\omega(A + Bt) - iB]e^{-i\omega t}, \tag{41}$$

and therefore vanishes on the physical shell $\omega^2 = 4$ for both Jordan coordinates.

Theorem 4.1 (Relative phase-space endomorphism)**.** *On the fixed compact background* (10)*, and before the final residual quotient, the identity map*

$$\iota : \mathcal{T}_{\mathrm{EM}}^{\mathrm{std}} \hookrightarrow \mathcal{T}_{\mathrm{WM}} \tag{42}$$

has nondegenerate target pullback:

$$\ker\left(\iota^* \Omega_{\mathrm{WM}}|_{\mathcal{T}_{\mathrm{EM}}^{\mathrm{std}}}\right) = 0. \tag{43}$$

The pullback is the block-orthogonal direct sum of (33)*,* (37)*,* (38)*, and* (40)*. Equivalently, R is $p_\lambda(M)$ on regular radiation, $4I$ on the physical $\ell = 1$ block, $I + N$ with $N^2 = 0$ and $\operatorname{rank} N = 2$ on the homogeneous block, and $-2I$ on each twist pair. The identity inclusion does not preserve the two-forms.*

Corollary 4.2 (Restricted linear Hamiltonians)**.** *On the restricted Einstein image, every linear Einstein–Maxwell Hamiltonian has a unique target-restricted Hamiltonian vector after application of the inverse blockwise relative operator.*

Remark 4.3. The corollary is not an embedding into the full target observable algebra. Extending a Hamiltonian away from the Einstein image and descending it under the final residual action are separate questions.

5 The complete generic axial extra branch

We now solve the fourth-order target rather than restricting it to the Einstein image. For generic $\ell \geq 2$, after local gauge contraction the axial coefficient order is

$$\Phi = (H_t, H_x, Q_t, Q_x)^T. \tag{44}$$

The contraction inverts only the harmless constant 2: neither k nor ω is inverted, so zero momentum is retained.

Put $s = \omega^2 - k^2$. The Einstein master polynomial and the extra polynomial are

$$q(s) = (s-\lambda)^2 - 2\lambda, \qquad p(s) = s - \lambda + \frac{2}{3}. \tag{45}$$

The fraction-field calculation over $\mathbb{Q}(\lambda, k)[\omega]$ is useful for discovery but, because it inverts k, cannot establish the zero-momentum specialization. We instead use

$$R_{\mathrm{phys}} = \mathbb{Q}[\lambda, k, \lambda^{-1}, (\lambda-2)^{-1}, (9\lambda-2)^{-1}] \tag{46}$$

and regard the reduced Hessian as a matrix over $R_{\mathrm{phys}}[\omega]$. In particular, k, ω, p, and q are not units.

There is a 2×2 unit minor $-\lambda^2$. Eliminating its corresponding block uses only the unit matrices $\operatorname{diag}(\lambda, -\lambda)$ and gives a Schur complement pT, where, after removing the common unit $-3/4$,

$$\begin{aligned}
&a = k^4 + 2k^2\lambda - k^2\omega^2 + \lambda^2 - 2\lambda,\\
&b = k\omega(k^2 + 2\lambda - \omega^2),\\
&d = k^2\omega^2 - \lambda^2 + 2\lambda\omega^2 + 2\lambda - \omega^4, \qquad T = -\frac{3}{4}\begin{pmatrix} a & b \\ b & d \end{pmatrix}.
\end{aligned} \tag{47}$$

The entries generate the unit ideal because

$$\omega^2(k^2 + 2\lambda - \omega^2)a - k\omega(k^2 + 2\lambda - \omega^2)b - \lambda(\lambda-2)d = \lambda^2(\lambda-2)^2. \tag{48}$$

It follows that the determinantal ideals are

$$I_1 = (1), \qquad I_2 = (1), \qquad I_3 = (p), \qquad I_4 = (p^2 q). \tag{49}$$

This is the no-k-torsion statement required for specialization.

For every physical fibre $\lambda = \ell(\ell+1)$, $\ell \geq 2$, and $k = 2\pi n/L$, $n \in \mathbb{Z}$, the resulting PID calculation over $K_{\ell,n}[\omega]$ has Smith factors

$$1, \quad 1, \quad p, \quad pq. \tag{50}$$

Moreover $\operatorname{Res}_\omega(p, q) = 4(9\lambda - 2)^2/81$ is nonzero on every physical fibre. The target module and its primary decomposition are therefore

$$\frac{K_{\ell,n}[\omega]}{(p)} \oplus \frac{K_{\ell,n}[\omega]}{(pq)} \cong \left(\frac{K_{\ell,n}[\omega]}{(p)}\right)^2 \oplus \frac{K_{\ell,n}[\omega]}{(q)}. \tag{51}$$

The axial Einstein master module is $K_{\ell,n}[\omega]/(q)$. Its inclusion is a module map, so its image is annihilated by q. Since q is a unit on each p-primary summand, that image lies entirely in the q-primary target summand. The quotient inclusion is injective by Theorem 2.2; both source and target q-primary summands have $K_{\ell,n}$-dimension $\deg_\omega q = 4$. The Einstein image therefore equals the complete q-primary summand. Taking the quotient in (51) gives

$$\mathcal{Q}^{\mathrm{extra}}_{\mathrm{ax},\ell,n} := (\mathcal{T}^{\mathrm{ax}}_{\mathrm{WM}}/\iota(\mathcal{T}^{\mathrm{ax}}_{\mathrm{EM}}))_{\ell,n} \cong \big(K_{\ell,n}[\omega]/(p)\big)^2. \tag{52}$$

No global unimodular Smith transformations over the multivariate ring are claimed. Equation (52), which is the theorem needed for the compact physical spectrum, includes $n = 0$.

Thus there are two independent additional cyclic summands on

$$\omega_e^2 = k^2 + \lambda - \frac{2}{3}. \tag{53}$$

Convenient representatives are

$$e_1 = \begin{pmatrix} -k^2 - \lambda \\ k\omega_e \\ \lambda \\ 0 \end{pmatrix}, \qquad e_2 = \begin{pmatrix} -k\omega_e \\ k^2 - \frac{2}{3} \\ 0 \\ \lambda \end{pmatrix}. \tag{54}$$

They are not defined by an arbitrary orthogonal complement: they represent a canonical quotient supported on a different invariant factor. At $k = 0$ the representatives reduce to columns $(-\lambda, 0, \lambda, 0)^T$ and $(0, -2/3, 0, \lambda)^T$; their lower 2×2 minor is λ^2, so the two extra directions remain independent. Two cyclic summands here mean four dimensions over $K_{\ell,n}$ before choosing a frequency root, two complex coefficients on the positive-frequency root, and four real phase-space dimensions per real spatial harmonic.

5.1 Direct Lee–Wald pairing

For any stationary positive-frequency block with time dependence $e^{-i\omega t}$, $\omega > 0$, define the normalized Hermitian current form

$$h_+(u, v) = \frac{\Omega_{\mathrm{WM}}(\bar{u}, v)}{-i\omega L N_{\ell m}}, \tag{55}$$

with the time orientation and overall action convention fixed by (3). A different positive normalization of the circle integral does not change its inertia. Reality and antisymmetry of the real Lee–Wald form give

$$\overline{h_+(u, v)} = \frac{\Omega_{\mathrm{WM}}(u, \bar{v})}{i\omega L N_{\ell m}} = \frac{\Omega_{\mathrm{WM}}(\bar{v}, u)}{-i\omega L N_{\ell m}} = h_+(v, u), \tag{56}$$

so h_+ is Hermitian on a common positive-frequency shell. In the basis (54) its Gram matrix G_X has entries

$$(G_X)_{11} = \frac{\lambda}{6}\left[9k^2\lambda^2 - 12k^2\lambda + 4k^2 + 9\lambda^3 - 18\lambda^2\right], \tag{57}$$

$$(G_X)_{12} = \frac{k\lambda\omega_e}{6}(3\lambda - 2)^2, \tag{58}$$

$$(G_X)_{22} = \frac{\lambda}{18}\left[27k^2\lambda^2 - 36k^2\lambda + 12k^2 + 36\lambda - 8\right]. \tag{59}$$

On the extra shell,

$$\det G_X = \frac{1}{3}\lambda^4(\lambda - 2)(9\lambda - 2) > 0. \tag{60}$$

The first principal minor is also positive for every real k and $\lambda \geq 6$. Hence inertia $G_X = (2, 0)$.

The Einstein and extra primary modules are mutually Lee–Wald orthogonal. The two Einstein master branches contribute one positive and one negative coefficient in the same direct-current convention. Therefore:

Theorem 5.1 (Axial target decomposition and current inertia)**.** *For every $\ell \geq 2$, every allowed compact momentum $k = 2\pi n/L$, $n \in \mathbb{Z}$, and before the final residual quotient, the complete generic axial Weyl–Maxwell solution fibre is the Lee–Wald orthogonal direct sum of the two Einstein master* *branches and the two cyclic extra summands* (52). *It is nondegenerate. On the positive-frequency eigenspace, the Hermitian current form* (55) *has inertia*

$$\operatorname{inertia}\big(h_+|_{\mathcal{H}_+^{\mathrm{ax}}}\big) = (3,1). \tag{61}$$

The extra block has inertia $(2,0)$*; the negative target direction lies on the minus Einstein-image branch.*

Remark 5.2 (What the sign does not prove). Theorem 5.1 is a sign theorem for a classical conserved current after a declared positive-frequency normalization. No compatible complex structure, positive one-particle inner product, BRST physical Hilbert space, or quantum time evolution has been constructed. Consequently the inertia $(3,1)$ is not, by itself, a quantum ghost or unitarity theorem. The negative target coefficient belongs to $\iota^*\Omega_{\mathrm{WM}}$, not to the independent Einstein–Maxwell source action. Familiar higher-derivative ghost conclusions [16] require additional quantum structure not constructed here.

5.2 A spectral coefficient extractor for the extra modes

Nondegeneracy makes the extra branch measurable within the declared reduced solution block. For an on-shell axial solution Φ, define

$$\mathcal{O}_X(\Phi) = G_X^{-1}\frac{\Omega_{\mathrm{WM}}(\bar{e},\Phi)}{-i\omega_e L N_{\ell m}} \in \mathbb{C}^2, \tag{62}$$

where $e = (e_1, e_2)^T$. The local Green identity gives slice independence, and direct substitution gives

$$\mathcal{O}_X(a_1e_1 + a_2e_2) = (a_1, a_2)^T, \qquad \mathcal{O}_X \circ \iota = 0 \tag{63}$$

on the complete generic axial Einstein image. Thus the additional solutions are neither a reduced-current radical nor a relabeling of Einstein modes. This is a conserved mode coefficient functional: it depends on the selected harmonic, stationary frequency splitting, and representatives. It is not a local spacetime observable. The extractor has not yet been descended through the final residual action or promoted to a relational, causal, asymptotic, Peierls, or quantum observable.

6 The normalization triangle

Higher-derivative currents are sensitive to integration-by-parts and action normalization conventions. Three independent constructions are therefore compared. First, the gauge-reduced axial equation operator is formally self-adjoint. In Fourier convention $\partial_t \mapsto -i\omega$, $\partial_x \mapsto ik$, its exact quadratic kernel in the order (H_t, H_x, Q_t, Q_x) is

$$K = \begin{pmatrix} -\frac{\lambda}{4}A & -\frac{k\lambda\omega}{4}B & \lambda & 0 \\ -\frac{k\lambda\omega}{4}B & \frac{\lambda}{4}C & 0 & -\lambda \\ \lambda & 0 & k^2+\lambda & k\omega \\ 0 & -\lambda & k\omega & -\lambda+\omega^2 \end{pmatrix}, \tag{64}$$

where

$$A = 3k^4 + 6k^2\lambda - 3k^2\omega^2 - 2k^2 + 3\lambda^2 - 3\lambda\omega^2 - 8\lambda + 6\omega^2, \tag{65}$$

$$B = 3k^2 + 3\lambda - 3\omega^2 + 4, \tag{66}$$

$$C = 3k^2\lambda - 3k^2\omega^2 - 6k^2 + 3\lambda^2 - 6\lambda\omega^2 - 8\lambda + 3\omega^4 + 2\omega^2. \tag{67}$$

It satisfies $K(\omega, k)^\dagger = K(-\omega, -k)^T$ and is the mixed Hessian of

$$S^{(2)}_{\mathrm{red}} = \frac{1}{2} \int \Phi(-\omega, -k)^T K(\omega, k) \Phi(\omega, k). \tag{68}$$

Second, the Green current generated by this Hessian obeys the exact local Green identity. Third, independent direct variation of the four-dimensional Weyl–Maxwell action gives a Lee–Wald current whose integral over S^2 equals $N_{\ell m}$ times that reduced Green current.

Proposition 6.1 (Three-point spectral interpolation)**.** *After the common harmonic norm $N_{\ell m}$ is removed, every entry of the difference between the direct integrated current and the reduced Green current is a polynomial in λ of degree at most two. It has no rational denominator or square-root dependence, and $SO(3)$ equivariance makes it independent of m. The direct independent-frequency calculations at $\lambda = 6, 12, 20$ vanish, so the difference vanishes for every $\lambda = \ell(\ell+1)$, $\ell \geq 2$, and every m.*

The polynomial bound is established before imposing either dispersion relation: the local curvature current contains at most two contracted sphere Laplacians after harmonic integration, and all integrations by parts use the polynomial identity $-\Delta_{S^2} Y = \lambda Y$. Thus no λ-dependent normalization is hidden in the interpolation. Covariant presymplectic potentials admit exact-improvement ambiguities [13]; here $a = A - \bar{A}$ is global on the fixed bundle, the Cauchy surface is closed, and integrated exact improvements vanish.

Proposition 6.2 (Hessian–Green–Lee–Wald triangle)**.** *In the generic axial block, the exact equation operator is the Hessian of the reconstructed quadratic reduced action; its local Green current equals the direct integrated four-dimensional Lee–Wald current with the normalization used in Theorem 5.1.*

Proposition 6.2 is sufficient for the linear phase-space and current-inertia theorems: it fixes the operator, local conservation law, direct four-dimensional current, and overall action sign independently. A literal second expansion of the unreduced four-dimensional action density would be a valuable additional audit, but it is not used or claimed here.

7 Local modes and global residual cohomology

Three complexes must not be conflated. First, after local Diff $\times U(1)$ reduction but before the final residual quotient, the metric-containing axial and polar modes (24), with their flux-forced Maxwell dressings, exist and are nonnull for both source and target currents. These are the exact compact representatives of ordinary gravitational radiation on this background.

Second, a later quotient by global conformal residual symmetries can have vanishing invariant one-particle cohomology on the closed cylinder. That is a statement about global invariant states in a different complex; it neither sets the preceding local-gauge-reduced solutions to zero nor guarantees that the spectral extractors (62) survive the last descent.

Third, $S^1 \times S^2$ has no null infinity. An asymptotically flat radiative phase space changes the allowed transformations and can turn boundary-acting transformations into charged symmetries. No conclusion about its scattering states follows from the compact residual quotient studied here.

Einstein is a sector, not a Weyl gauge. The exact relation established here is

$$\mathrm{Sol}^{\mathrm{lin,std}}_{\mathrm{EM}} \subsetneq \mathrm{Sol}^{\mathrm{lin,ax+standard}}_{\mathrm{WM}} \tag{69}$$

on the declared background and harmonic strata. The strictness is witnessed by (52). Boundary conditions may later select an Einstein sector, but such selection is additional dynamics plus boundary data, not a local Weyl gauge fixing.

8 Open gates and theorem boundary

The following are not prerequisites for the scoped axial paper, but they are required to widen its title-level theorem.

1. **Polar extra branch.** Repeat the exact operator/module, Lee–Wald, radical, sign, and detector analysis in even parity. Only then is the complete additional compact target parity-classified.

2. **Final residual descent.** Determine which standard and extra coefficient observables survive the declared global residual quotient. Until then physical mode means the local-gauge-reduced compact phase space, not the final residual state space.

3. **Literal action-density expansion.** This is an independent normalization audit. The present theorem instead uses the exact Hessian–Green–direct-Lee–Wald triangle of Proposition 6.2.

4. **Nonlinear extension.** Linear inclusion does not imply that an Einstein tangent extends to a Weyl–Maxwell family. Fixed-bundle Taub tests already show scoped second-order obstructions; those belong to a separate nonlinear charge-fibre paper. Selected algebraically special Kundt waves on direct-product electrovacua are known to survive broad higher-order modifications [8, 9]; the open question here is whether the complete harmonic Einstein tangent, rather than those special exact families, is nonlinearly closed in this fixed compact theory.

5. **Causal boundary selection and scattering.** A future theorem must specify asymptotically flat function spaces, retarded/advanced complexes, radiative symplectic flux, residual charges, and whether the Einstein subspace is dynamically closed. Bondi energy, black holes, and scattering are not inferred here.

The strongest present conclusion is therefore:

> On the fixed compact product, ordinary Einstein–Maxwell waves embed exactly and nondegenerately in the linear Weyl–Maxwell phase space, but with a different Hamiltonian form. Weyl–Maxwell theory also has a complete generic axial branch of genuine, nonradical additional solutions. Neither branch has yet been classified after the final residual quotient or as an asymptotic quantum particle space.

9 Exact certification and reproducibility

Every theorem in this manuscript is a synthesis of exact machine-readable certificates. The companion claim map `paper/10-compact-einstein-maxwell-weyl-phase-space-claim-map.json` pins the full SHA-256 hashes and fails closed if an input changes.

The scoped verification commands are listed below; the breakable path formatting is typographical and each line is one shell command.

```
python3bridge/einstein_sector/verify_einstein_maxwell_product_incidence.py
python3bridge/einstein_sector/verify_einstein_maxwell_chevreton_tangent.py
python3bridge/einstein_sector/verify_einstein_maxwell_chevreton_formal_linearization.py
python3bridge/einstein_sector/verify_einstein_maxwell_radiative_symplectic_matching.py
python3bridge/einstein_sector/verify_einstein_maxwell_weyl_standard_harmonic_symplectic_inclusion.
py
python3bridge/einstein_sector/verify_einstein_maxwell_weyl_axial_operator.py
python3bridge/einstein_sector/verify_einstein_maxwell_weyl_axial_physical_ring.py
python3bridge/einstein_sector/verify_einstein_maxwell_weyl_axial_lee_wald_completion.py
```

Result	Certificate role
EINSTEIN_MAXWELL_PRODUCT_INCIDENCE	Common exact product background and fixed-flux rational fixture.
EINSTEIN_MAXWELL_CHEVRETON_TANGENT	Complete linear on-shell solution inclusion.
EINSTEIN_MAXWELL_CHEVRETON_FORMAL_LINEARIZATION	Dual-number bridge from the on-shell derivation to every formal Jacobi field, without an integrability assumption.
COMPACT_EM_RADIATIVE_SYMPLECTIC_MATCHING	Positive source radiative pairing and direct Lee–Wald match.
EINSTEIN_MAXWELL_WEYL_STANDARD_HARMONIC_SYMPLECTIC_INCLUSION	Complete standard image, mixed orthogonality, and nondegenerate pullback.
EINSTEIN_MAXWELL_WEYL_AXIAL_OPERATOR	Generic target operator and fraction-field invariant factors.
EINSTEIN_MAXWELL_WEYL_AXIAL_PHYSICAL_RING	Physical-ring Fitting ideals and every compact-momentum fibre, including $k = 0$.
EINSTEIN_MAXWELL_WEYL_AXIAL_LEE_WALD_COMPLETION	Direct four-dimensional current and complete axial current inertia.
EINSTEIN_MAXWELL_WEYL_AXIAL_REDUCED_ACTION_HESSIAN	Reduced Hessian and normalization triangle.
EINSTEIN_MAXWELL_WEYL_AXIAL_EXTRA_DETECTOR	Conserved extra spectral coefficient extractors before residual descent.

Table 2: Principal exact inputs. The claim map also pins the exceptional and mixed-block component certificates.

`python3bridge/einstein_sector/verify_einstein_maxwell_weyl_axial_reduced_action_hessian.py`
`python3bridge/einstein_sector/verify_einstein_maxwell_weyl_axial_extra_detector.py`
`python3bridge/einstein_sector/verify_compact_linear_paper_claim_map.py` The manuscript itself is checked with two passes of `pdflatex`.

10 Conclusion

The Einstein sector is visible without interpretive guesswork. On a common compact Einstein–Maxwell/Weyl–Maxwell background, the complete standard Einstein–Maxwell harmonic tangent injects into the Weyl–Maxwell solution space and remains nondegenerate under the target Lee–Wald form. The identity inclusion nevertheless changes that form in every class of blocks: by branch-dependent weights on regular radiation, a factor four on the physical $\ell = 1$ quotient, a unipotent shear on the homogeneous sector, and a factor minus two on the twists.

The target is strictly larger. Its generic axial quotient contains two additional nonradical solution summands, detected by conserved coefficient extractors, and the normalized positive-frequency axial current has inertia $(3, 1)$. These statements settle where ordinary waves and one genuine extra branch sit before the final residual quotient. They do not turn a classical sign into a quantum ghost theorem, and they do not replace the separate causal problem of selecting an Einstein scattering sector at infinity.

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